

JT/T808 Communication Protocol (2013 Version)

ELOCK Extended Version (A13)

Revision History

Date	Version	Description	Author
2016-03-18	A1	First version after re-merging and archiving	Yuan
2016-03-23	A2	Added text message upload protocol; Added online upgrade identifier to GPS status bit	Yuan
2016-06-02	A3	Added dedicated authorization command; Added version number upload Additional Information; Added text message command description (Appendix 4)	Yuan
2016-12-02	A4	Added local password modification command	Yuan
2016-12-22	A5	Added IC card related protocol (only for devices with IC card function)	Yuan
2017-03-02	A6	Added door magnetic switch protocol (yellow bottom part) (Table 20 Definition of Additional Information 0xEE+10.2.3.2) (only for devices with door magnetic function)	Yuan
2017-11-21	A7	Registration information changed from IMSI to ICCID	Yuan
2017-12-22	A8	TT extension modification parts marked in blue font, titles in blue indicate the main text is TT extension	Yuan
2018-11-19	A9	Added 4G version LBS positioning extension	Yuan
2018-12-29	A10	Modified lock operation reply, added LINECODE operation identifier; Added IC card blacklist command	Yuan
2018-1-28	A11	Added tilt alarm, Gsensor acceleration upload; Fixed "query terminal attributes" reply Typo: change 0200 to 0201	Yuan
2018-2-13	A12	Deleted E5 lock status 2 extension, deleted Appendix 4; Modified E7 expansion alarm definition; "Terminal mobile number" description changed to "Terminal ID number"	Yuan
2018-5-6	A13	Modified operation identifier description, upload operation password	Yuan

1 Foreword

This specification specifies the communication protocol and data format between road transport vehicle satellite positioning system BeiDou compatible vehicle terminals and **TT electronic lock devices** (hereinafter referred to as terminals) and supervision/monitoring platforms (hereinafter referred to as platforms), including protocol basics, communication connection, message processing, protocol classification and description, and data format.

This specification applies to communication between TT terminals and platforms.

Yellow bottom part: parts that need to be done for accessing TT devices, or content that developers need to read

Blue font: TT modified and extended parts, also content that needs to be read and implemented

2 Normative References

The following documents are essential for the application of this document. For dated references, only the version dated applies. For undated references, the latest version (including all amendments) applies. GB/T 2260 Codes of the People's Republic of China for administrative divisions; GB/T 19056 Vehicle traveling data recorder; JT/T 415-2006 Road transport electronic government affairs platform - coding rules; JT/T 794 Road transport vehicle satellite positioning system - vehicle terminal technical requirements

3 Terms, Definitions and Abbreviations

3.1 Terms and Definitions

The following terms and definitions apply to this document.

3.1.1 abnormal data communication link Wireless communication link disconnection, or temporary suspension (during a call).

3.1.2 register The terminal sends a message to the platform to inform that it is installed on a certain vehicle.

3.1.3 unregister The terminal sends a message to the platform to inform that it has been removed from the installed vehicle.

3.1.4 authentication After the terminal connects to the platform, it sends a message to the platform so that the platform can verify its identity.

3.1.5 location reporting strategy Timed, distance-based reporting, or a combination of both.

3.1.6 location reporting program Rules for determining the interval of periodic reporting based on relevant conditions.

3.1.7 additional points report while turning The terminal sends location information reporting messages when it determines that the vehicle is turning. The sampling frequency is not lower than 1Hz, the vehicle azimuth change rate is not lower than 15°/s, and it lasts for at least 3s.

3.1.8 answering strategy Rules for the terminal to automatically or manually answer incoming calls.

3.1.9 SMS text alarm When the terminal alarms, it sends text information via SMS.

3.1.10 event item Event items are preset by the platform to the terminal, consisting of event code and event name. The driver operates the terminal when encountering the corresponding event, triggering event reporting to the platform.

3.2 Abbreviations

The following abbreviations apply to this document.

APN — Access Point Name

GZIP — A GNU free software file compression program (GNUZip)

LCD — Liquid Crystal Display

RSA — An asymmetric cryptographic algorithm (developed by Ron Rivest, Adi Shamir, Len Adleman, named after the three)

SMS — Short Message Service

TCP — Transmission Control Protocol

TTS — Text to Speech

UDP — User Datagram Protocol

VSS — Vehicle Speed Sensor

4 Protocol Basics

4.1 Communication Method

The communication method adopted by the protocol shall comply with the relevant provisions in JT/T 794. The communication protocol adopts TCP or UDP, with the platform as the server and the terminal as the client. When the data communication link is abnormal, the terminal can communicate via SMS messages.

Note: TT devices adopt TCP protocol (socket).

4.2 Data Types

The data types used in protocol messages are shown in Table 1: Table 1 Data Types

Data Type	Description and Requirements
BYTE	Unsigned single-byte integer (byte, 8 bits)
WORD	Unsigned double-byte integer (word, 16 bits)
DWORD	Unsigned four-byte integer (double word, 32 bits)
BYTE[n]	n bytes
BCD[n]	8421 code, n bytes
STRING	GBK encoding, if no data, leave empty

4.3 Transmission Rules

The protocol adopts big-endian network byte order to transmit words and double words.

Agreements are as follows:

- Byte (BYTE) transmission convention: transmitted according to byte stream;
- Word (WORD) transmission convention: transmit the high eight bits first, then the low eight bits;
- Double word (DWORD) transmission convention: transmit the high 24 bits first, then the high 16 bits, then the high eight bits, and finally the low eight bits.

4.4 Message Composition

4.4.1 Message Structure

Each message consists of a marker bit, message header, message body, and check code. The message structure is shown in Figure 1:

Marker Bit	Message Header	Message Body	Check Code	Marker Bit
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Figure 1 Message Structure Diagram

4.4.2 Marker Bit

0x7e is used. If 0x7e appears in the check code, message header, and message body, escape processing is required. The escape rules are defined as follows: 0x7e <—> 0x7d followed by 0x02; 0x7d <—> 0x7d followed by 0x01.

The escape processing process is as follows:

When sending a message: message encapsulation —> calculate and fill check code —> escape;

When receiving a message: escape restoration —> verify check code —> parse message.

Example:

Send a data packet with content 0x30 0x7e 0x08 0x7d 0x55, after encapsulation: 0x7e 0x30 0x7d 0x02 0x08 0x7d 0x01 0x55 0x7e.

4.4.3 Message Header

The message header content is shown in Table 2: Table 2 Message Header Content

Starting Byte	Field	Data Type	Description and Requirements
0	Message ID	WORD	Command word
2	Message Body Properties	WORD	Message body attribute format structure is shown in Figure 2
4	Terminal ID Number	BCD[6]	Device ID, parsed as 12 digits, leading 0s can be omitted. Example: 0x023456789012 ==> 23456789012
10	Message Serial Number	WORD	Cyclically accumulates from 0 in sending order
12	Message		If the relevant identifier bit in the

	Package Item		message body properties determines message sub-packet processing, this item has content; otherwise there is no such item (TT devices generally do not have this item)
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The message body attribute format structure is shown in Figure 2:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		Sub-packet	Data Encryption			Message Body Length									

Figure 2 Message Body Attribute Format Structure Diagram

Data encryption method: —— bit10-bit12 are data encryption identifier bits;

—— When all three bits are 0, it means the message body is not encrypted;

—— When bit 10 is 1, it means the message body is encrypted by RSA algorithm;

—— Others reserved.

Sub-packet: When bit 13 in the message body properties is 1, it means the message body is a long message and is processed by sub-packet sending. The specific sub-packet information is determined by the message package item; if bit 13 is 0, there is no message package item field in the message header. The message package item content is shown in Table 3:

Table 3 Message Package Item Content

Starting Byte	Field	Data Type	Description and Requirements
0	Total Message Packets	WORD	Total number of packets after message sub-packeting
2	Packet Sequence Number	WORD	Starting from 1

4.4.4 Check Code

The check code starts from the message header, XORs byte by byte with the same position, until the byte before the check code, occupying one byte.

5 Communication Connection

5.1 Connection Establishment

The daily data connection between the terminal and the platform can adopt TCP or UDP. After the terminal resets, it should establish a connection with the platform as soon as possible. After the connection is established, immediately send the terminal authentication message to the platform for authentication.

5.2 Connection Maintenance

After the connection is established and the terminal authentication is successful, in the absence of normal data packet transmission, the terminal should periodically send terminal heartbeat

messages to the platform. After receiving, the platform sends a platform universal response message to the terminal. The sending period is specified by terminal parameters.

5.3 Connection Disconnection

Both the platform and the terminal can actively disconnect according to the TCP protocol. Both parties should actively determine whether the TCP connection is disconnected.

Methods for the platform to determine TCP connection disconnection:

- Determine that the terminal actively disconnects according to the TCP protocol;
- A terminal with the same identity establishes a new connection, indicating that the original connection is disconnected;
- No message sent by the terminal is received within a certain period of time, such as terminal heartbeat.

Methods for the terminal to determine TCP connection disconnection:

- Determine that the platform actively disconnects according to the TCP protocol;
- Data communication link disconnection;
- The data communication link is normal, but no response is received after reaching the number of retransmissions.

6 Message Processing

6.1 TCP and UDP Message Processing

6.1.1 Platform-initiated Messages

All platform-initiated messages require terminal response. The response is divided into universal response and dedicated response, determined by each specific function protocol. After the sender waits for response timeout, it should retransmit the message. The response timeout time and number of retransmissions are specified by platform parameters. The calculation formula for the response timeout time after each retransmission is shown in formula (1):

$$TN+1 = TN \times (N+1) \dots\dots\dots (1)$$

Where:

TN+1 — response timeout time after each retransmission;

TN — previous response timeout time;

N — number of retransmissions.

6.1.2 Terminal-initiated Messages

6.1.2.1 Normal Data Communication Link

When the data communication link is normal, all terminal-initiated messages require platform response. The response is divided into universal response and dedicated response, determined by each specific function protocol. After the terminal waits for response timeout, it should retransmit the message. The response timeout time and number of retransmissions are specified by terminal

parameters. The response timeout time after each retransmission is calculated by formula (1). For key alarm messages sent by the terminal, if no response is received after reaching the number of retransmissions, they should be saved. Thereafter, before sending other messages, the saved key alarm messages should be sent first.

6.1.2.2 Abnormal Data Communication Link

When the data communication link is abnormal, the terminal should save the location information report messages that need to be sent. After the data communication link returns to normal, immediately send the saved messages.

8 Data Format

8.1 Terminal Universal Response

Message ID: 0x0001. The terminal universal response message body data format is shown in Table 4.

Table 4 Terminal Universal Response Message Body Data Format

Starting Byte	Field	Data Type	Description and Requirements
0	Response Serial Number	WORD	Serial number of the corresponding platform message
2	Response ID	WORD	ID of the corresponding platform message
4	Result	BYTE	0: Success/confirmed; 1: Failure; 2: Message error; 3: Not supported

8.2 Platform Universal Response

Message ID: 0x8001. The platform universal response message body data format is shown in Table 5.

Table 5 Platform Universal Response Message Body Data Format

Starting Byte	Field	Data Type	Description and Requirements
0	Response Serial Number	WORD	Serial number of the corresponding terminal message
2	Response ID	WORD	ID of the corresponding terminal message
4	Result	BYTE	0: Success/confirmed; 1: Failure; 2: Message error; 3: Not supported; 4: Alarm processing confirmed

8.3 Terminal Heartbeat

Message ID: 0x0002. The terminal heartbeat data message body is empty.

8.5 Terminal Registration

Message ID: 0x0100.

The terminal registration message body data format is shown in Table 7.

Table 7 Terminal Registration Message Body Data Format (Applicable to TT devices)

Starting Byte	Field	Data Type	Description and Requirements
0	ICCID	BYTE	20-byte ICCID

Note: Registration information is sent only when authentication fails.

8.6 Terminal Registration Response

Message ID: 0x8100.

The terminal registration response message body data format is shown in Table 8.

Table 8 Terminal Registration Response Message Body Data Format

Starting Byte	Field	Data Type	Description and Requirements
0	Response Serial Number	WORD	Serial number of the corresponding terminal registration message
2	Result	BYTE	0: Success; 1: Vehicle already registered; 2: No such vehicle in database; 3: Terminal already registered; 4: No such terminal in database
3	Authentication Code	STRING	This field exists only after successful authentication

8.8 Terminal Authentication

Message ID: 0x0102.

The terminal authentication message body data format is shown in Table 9.

Table 9 Terminal Authentication Message Body Data Format

Starting Byte	Field	Data Type	Description and Requirements
0	Authentication Code	STRING	Terminal reports authentication code after reconnection

Notes:

- 1: For TT terminals that have not been issued with an authentication code, the authentication code content is SIM card ICCID (20 bytes) + device version number.
- 2: 0100, 0102 commands are sent only in real-time mode, not in sleep mode.

8.9 Set Terminal Parameters

Message ID: 0x8103.

The set terminal parameters message body data format is shown in Table 10.

Table 10 Terminal Parameters Message Body Data Format

Starting Byte	Field	Data Type	Description and Requirements
0	Total Parameters	BYTE	
1	Parameter Item List		Parameter item format is shown in Table 11

Table 11 Terminal Parameter Item Data Format

Field	Data Type	Description and Requirements
Parameter ID	DWORD	Parameter ID definition and description are shown in Table 12
Parameter Length	BYTE	
Parameter Value		If it is a multi-value parameter, multiple parameters with the same ID are used in the message, such as dispatch center phone number

Table 12 Terminal Parameter Definitions and Descriptions (Note: Yellow bottom part are parameters used by TT devices)

Parameter ID	Data Type	Description and Requirements
0x0001	DWORD	Terminal heartbeat sending interval, in seconds (s)
0x0002	DWORD	TCP message response timeout time, in seconds (s)
0x0003	DWORD	TCP message retransmission count
0x0004	DWORD	UDP message response timeout time, in seconds (s)
0x0005	DWORD	UDP message retransmission count
0x0006	DWORD	SMS message response timeout time, in seconds (s)
0x0007	DWORD	SMS message retransmission count
0x0008-0x000F		Reserved
0x0010	STRING	SIM card APN
0x0011	STRING	SIM card dial-up username
0x0012	STRING	SIM card dial-up password
0x0013	STRING	Main server address, IP or domain name
0x0014	STRING	Backup server APN, wireless communication dial-up access point
0x0015	STRING	Backup server wireless communication dial-up username
0x0016	STRING	Backup server wireless communication dial-up password
0x0017	STRING	Backup server address, IP or domain name
0x0018	DWORD	Server TCP port (suggest 4-digit port)
0x0019	DWORD	Server UDP port
0x001A	STRING	Road transport certificate IC card authentication main server IP address or domain name
0x001B	DWORD	Road transport certificate IC card authentication

		main server TCP port
0x001C	DWORD	Road transport certificate IC card authentication main server UDP port
0x001D	STRING	Road transport certificate IC card authentication backup server IP address or domain name, port same as main server
0x001E-0x001F		Reserved
0x0020	DWORD	Location reporting strategy: 0: Timed reporting; 1: Distance-based reporting; 2: Timed and distance-based reporting
0x0021	DWORD	Location reporting scheme: 0: According to ACC status; 1: According to login status and ACC status, first judge login status, if logged in then according to ACC status
0x0022	DWORD	Driver not logged in reporting time interval, in seconds (s), >0
0x0023-0x0026	DWORD	Reserved
0x0027	DWORD	Sleep time reporting time interval, in seconds (s), >0. =0 means cancel timed sleep, >0 means enable timed sleep
0x0028	DWORD	Emergency alarm time reporting time interval, in seconds (s), >0
0x0029	DWORD	Power-saving time reporting time interval, in seconds (s), >0
0x002A	BCD[6]	TT time extension: 6BYTE time, hexadecimal literal meaning: e.g. 0x080402152050 is 2008-4-2 15:20:50
0x002B	DWORD	Reserved
0x002C	DWORD	Default distance reporting interval, in meters (m), >0
0x002D	DWORD	Driver not logged in reporting distance interval, in meters (m), >0
0x002E	DWORD	Sleep time reporting distance interval, in meters (m), >0
0x002F	DWORD	Emergency alarm time reporting distance interval, in meters (m), >0
0x0030	DWORD	Turn supplementary angle, <180
0x0031	WORD	Electronic fence radius (illegal displacement threshold), in meters
0x0032-0x003F		Reserved
0x0040	STRING	Monitoring platform phone number
0x0041	STRING	Reset phone number, can use this number to dial the terminal phone to let terminal reset
0x0042	STRING	Restore factory settings phone number, can use this number to dial terminal phone to let terminal restore

		factory settings
0x0043	STRING	Monitoring platform SMS phone number
0x0044	STRING	Receive terminal SMS text alarm number
0x0045	DWORD	Terminal phone answering strategy: 0: Auto answer; 1: Auto answer when ACC ON, manual when OFF
0x0046	DWORD	Maximum call duration each time, in seconds (s), 0 means no call allowed, 0xFFFFFFFF means no limit
0x0047	DWORD	Maximum call duration per month, in seconds (s), 0 means no call allowed, 0xFFFFFFFF means no limit
0x0048	STRING	Monitoring phone number
0x0049	STRING	Supervision platform SMS short number
0x004A-0x004F		Reserved
0x0050	DWORD	Alarm shield word, corresponds to the alarm flag bit in the location information report, the corresponding bit is 1 then the corresponding alarm is shielded
0x0051	DWORD	Alarm send text SMS switch, corresponds to the alarm flag bit in location information report, corresponding bit is 1 then corresponding alarm sends text SMS
0x0052	DWORD	Alarm shooting switch, corresponds to the alarm flag bit in location information report, corresponding bit is 1 then corresponding alarm triggers camera shooting
0x0053	DWORD	Alarm shooting storage flag, corresponds to the alarm flag bit in location information report, corresponding bit is 1 then photos taken during corresponding alarm are stored, otherwise uploaded in real-time
0x0054	DWORD	Key alarm flag, corresponds to the alarm flag bit in location information report, corresponding bit is 1 then corresponding alarm is key alarm
0x0055	DWORD	Maximum speed, in km/h
0x0056	DWORD	Overspeed duration, in seconds (s)
0x0057	DWORD	Continuous driving time threshold, in seconds (s)
0x0058	DWORD	Cumulative driving time threshold of the day, in seconds (s)
0x0059	DWORD	Minimum rest time, in seconds (s)
0x005A	DWORD	Maximum parking time, in seconds (s)
0x005B	WORD	Overspeed driving alarm difference, in 1/10Km/h
0x005C	WORD	Fatigue driving alarm difference, in seconds (s), >0
0x005D	WORD	Collision alarm parameter setting: b7-b0: collision time, unit 4ms; b15-b8: collision acceleration, unit 0.1g, setting range: 0-79, default 10
0x005E	WORD	Roll-over alarm parameter setting: roll-over angle, unit 1 degree, default 30 degrees

0x005F-0x0063		Reserved
0x0064	DWORD	Timed shooting control, see Table 13
0x0065	DWORD	Distance-based shooting control, see Table 14
0x0066-0x006F		Reserved
0x0070	DWORD	Image/video quality, 1-10, 1 is best
0x0071	DWORD	Brightness, 0-255
0x0072	DWORD	Contrast, 0-127
0x0073	DWORD	Saturation, 0-127
0x0074	DWORD	Chromaticity, 0-255
0x0075-0x007F		Reserved
0x0080	DWORD	Vehicle odometer reading, 1/10km
0x0081	WORD	Province ID where the vehicle is located
0x0082	WORD	City ID where the vehicle is located
0x0083	STRING	Motor vehicle license plate number issued by public security traffic management department
0x0084	BYTE	License plate color, according to JT/T415-2006 5.4.12
0x0090	BYTE	GNSS positioning mode, defined as follows: bit0, 0: GPS positioning disabled, 1: GPS positioning enabled; bit1, 0: BeiDou positioning disabled, 1: BeiDou positioning enabled; bit2, 0: GLONASS positioning disabled, 1: GLONASS positioning enabled; bit3, 0: Galileo positioning disabled, 1: Galileo positioning enabled
0x0091	BYTE	GNSS baud rate, defined as follows: 0x00: 4800; 0x01: 9600; 0x02: 19200; 0x03: 38400; 0x04: 57600; 0x05: 115200
0x0092	BYTE	GNSS module detailed positioning data output frequency, defined as follows: 0x00: 500ms; 0x01: 1000ms (default); 0x02: 2000ms; 0x03: 3000ms; 0x04: 4000ms
0x0093	DWORD	GNSS module detailed positioning data acquisition frequency, in seconds, default 1
0x0094	BYTE	GNSS module detailed positioning data upload method: 0x00, local storage, no upload (default); 0x01, upload by time interval; 0x02, upload by distance interval; 0x0B, upload by time interval, automatically stop after reaching transmission time; 0x0C, upload by distance interval, automatically stop after reaching distance; 0x0D, upload by count, automatically stop after reaching upload count
0x0095	DWORD	GNSS module detailed positioning data upload setting: when upload method is 0x01, unit is seconds;

		when upload method is 0x02, unit is meters; when upload method is 0x0B, unit is seconds; when upload method is 0x0C, unit is meters; when upload method is 0x0D, unit is count
0x0100	DWORD	CAN bus channel 1 acquisition time interval (ms), 0 means no acquisition
0x0101	WORD	CAN bus channel 1 upload time interval (s), 0 means no upload
0x0102	DWORD	CAN bus channel 2 acquisition time interval (ms), 0 means no acquisition
0x0103	WORD	CAN bus channel 2 upload time interval (s), 0 means no upload
0x0110	BYTE[8]	CAN bus ID individual acquisition setting: bit63-bit32 indicates this ID acquisition time interval (ms), 0 means no acquisition; bit31 CAN channel number, 0: CAN1, 1: CAN2; bit30 indicates frame type, 0: standard frame, 1: extended frame; bit29 data acquisition method, 0: original data, 1: calculated value of acquisition interval; bit28-bit0 indicates CAN bus ID
0x0111-0x01FF	BYTE[8]	Used for other CAN bus ID individual acquisition settings
0xF000-0xFFFF		User-defined

Note: 1. Please send the 0x002A setting time command 500ms after receiving the device authentication and successful authentication, to calibrate the device time.

8.10 Query Terminal Parameters

Message ID: 0x8104. The query terminal parameters message body is empty.

8.12 Query Terminal Parameters Response

Message ID: 0x0104.

The query terminal parameters response message body data format is shown in Table 16.

Table 16 Query Terminal Parameters Response Message Body Data Format

Starting Byte	Field	Data Type	Description and Requirements
0	Response Serial Number	WORD	Serial number of the corresponding terminal parameter query message
2	Response Parameter Count	BYTE	
3	Parameter Item List		Parameter item format and definition see Table 10

8.13 Terminal Control

Message ID: 0x8105.

The terminal control message body data format is shown in Table 17.

Table 17 Terminal Control Message Body Data Format

Starting Byte	Field	Data Type	Description and Requirements
0	Command Word	BYTE	Terminal control command word description see Table 18
1	Command Parameter	STRING	Command parameter format is described later, each field is separated by semicolon ";", each STRING field is first GBK encoded and then composed into the message

Table 18 Terminal Control Command Word Description

Command Word	Command Parameter	Description and Requirements
1	Command parameter format see Table 19	Wireless upgrade. Parameters are separated by semicolons. Command is as follows: "URL address;dial-up point name;dial-up username;dial-up password;TCP port;UDP port;manufacturer ID;hardware version;firmware version;connection to specified server timeout"
2	Command parameter format see Table 19	Control terminal to connect to specified server. Parameters are separated by semicolons. Control command is as follows: "connection control;supervision platform authentication code;dial-up point name;dial-up username;dial-up password;address;TCP port;UDP port;connection to specified server timeout", if a parameter has no value, leave it empty; if connection control value is 1, then no subsequent parameters
3	None	Terminal power off
4	None	Terminal reset
5	None	Terminal restore factory settings
6	None	Close data communication
7	None	Close all wireless communication

Table 19 Command Parameter Format

Field	Data Type	Description and Requirements
Connection Control	BYTE	0: Switch to specified supervision platform server, after connecting to this server enter emergency state, in this state the supervision platform that issues the control command can send control commands including SMS; 1: Switch back to original provincial supervision

		platform server and restore normal state
Dial-up Point Name	STRING	Generally server APN, wireless communication dial-up access point, if network system is CDMA, this value is PPP connection dial-up number
Dial-up Username	STRING	Server wireless communication dial-up username
Dial-up Password	STRING	Server wireless communication dial-up password
Address	STRING	Server address, IP or domain name
TCP Port	WORD	Server TCP port
UDP Port	WORD	Server UDP port
Manufacturer ID	BYTE[5]	Terminal manufacturer code
Supervision Platform Authentication Code	STRING	Authentication code issued by the supervision platform, only used for terminal authentication after connecting to supervision platform, terminal connects back to original platform and restores original authentication code
Hardware Version	STRING	Terminal hardware version number, determined by manufacturer
Firmware Version	STRING	Terminal firmware version number, determined by manufacturer
URL Address	STRING	Complete URL address
Connection to Specified Server Timeout	WORD	Unit: minutes (min), value 0 means after the terminal receives the upgrade or connects to specified server command, before the validity period expires, the terminal should connect back to the original address. If value is 0, it means always connect to the specified server

8.14 Query Terminal Attributes

Message ID: 0x8107. The query terminal attributes message body is empty.

TT Note: For convenience and flexibility, TT devices reply in the form of 0201+extended information

8.17 Terminal Upgrade Result Notification

Message ID: 0x0108. After the terminal completes upgrade and reconnects, use this command to notify the monitoring center. The terminal upgrade result notification message body data format is shown in Table 22.

Table 22 Terminal Upgrade Result Notification Message Body Data Format

Starting Byte	Field	Data Type	Description and Requirements
0	Upgrade Type	BYTE	0: Terminal; 12: Road transport certificate IC card reader; 52: BeiDou

			satellite positioning module
1	Upgrade Result	BYTE	0: Success; 1: Failure; 2: Cancel

8.18 Location Information Report

This entire section is mandatory

Message ID: 0x0200. The location information report message body consists of basic location information and a list of location additional information items. The message structure is shown in Figure 3:

Basic Location Information	Location Additional Information Item List
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Figure 3 Location Report Message Structure Diagram

The location additional information item list is composed of each location additional information item, or may be absent, determined according to the length field in the message header.

The basic location information data format is shown in Table 23.

Table 23 Basic Location Information Data Format

Starting Byte	Field	Data Type	Description and Requirements
0	Alarm Flag	DWORD	Alarm flag bit definition see Table 24
4	Status	DWORD	Status bit definition see Table 25
8	Latitude	DWORD	Latitude value in degrees multiplied by 10 to the power of 6, accurate to one millionth of a degree
12	Longitude	DWORD	Longitude value in degrees multiplied by 10 to the power of 6, accurate to one millionth of a degree
16	Altitude	WORD	Altitude, in meters (m)
18	Speed	WORD	1/10km/h
20	Direction	WORD	0-359, north is 0, clockwise
21	Time	BCD[6]	YY-MM-DD-hh-mm-ss (GMT+8 time, all times involved after this standard adopt this time zone)

Table 24 Alarm Flag Bit Definition

Bit	Definition	Processing Description
0	1: Emergency alarm, triggered after touching the alarm switch	Cleared after receiving response
1	1: Overspeed alarm	Flag maintained until alarm condition is release
2	1: Fatigue driving	Flag maintained until alarm condition is release

3	1: Danger warning	Cleared after receiving response
4	1: GNSS module failure	Flag maintained until alarm condition is release
5	1: GNSS antenna not connected or cut	Flag maintained until alarm condition is release
6	1: GNSS antenna short circuit	Flag maintained until alarm condition is release
7	1: Terminal main power undervoltage	Flag maintained until alarm condition is release
8	1: Terminal main power power off	Flag maintained until alarm condition is release
9	1: Terminal LCD or display failure	Flag maintained until alarm condition is release
10	1: TTS module failure	Flag maintained until alarm condition is release
11	1: Camera failure	Flag maintained until alarm condition is release
12	1: Road transport certificate IC card module failure	Flag maintained until alarm condition is release
13	1: Overspeed warning	Flag maintained until alarm condition is release
14	1: Fatigue driving warning	Flag maintained until alarm condition is release
15-17	Reserved	
18	1: Cumulative driving overtime of the day	Flag maintained until alarm condition is release
19	1: Overtime parking	Flag maintained until alarm condition is release
20	1: Enter/exit area	Cleared after receiving response
21	1: Enter/exit route	Cleared after receiving response

22	1: Road segment driving time insufficient/too long	Cleared after receiving response
23	1: Route deviation alarm	Flag maintained until alarm condition is release
24	1: Vehicle VSS failure	Flag maintained until alarm condition is releaser
25	1: Vehicle fuel level abnormal	Flag maintained until alarm condition is release
26	1: Vehicle stolen (through vehicle anti-theft device)	Flag maintained until alarm condition is release
27	1: Vehicle illegal ignition	Cleared after receiving response
28	1: Vehicle illegal displacement	Cleared after receiving response
29	1: Collision warning	Flag maintained until alarm condition is release
30	1: Roll-over warning	Flag maintained until alarm condition is clear
31	1: Illegal door opening alarm (when terminal does not set area, do not judge illegal door opening)	Cleared after receiving response

Note: When an alarm or warning occurs, immediately report location information

Table 25 Status Bit Definition Note: When status changes, immediately report location information

Bit	Status
0	0: ACC off; 1: ACC on
1	0: Not positioned; 1: Positioned
2	0: North latitude; 1: South latitude
3	0: East longitude; 1: West longitude
4	0: Operating status; 1: Out-of-service status
5	0: Latitude and longitude not encrypted by confidentiality plug-in; 1: Latitude and longitude encrypted by confidentiality plug-in
6-7	Reserved
8-9	00: Empty vehicle; 01: Half load; 10: Reserved; 11: Full load (can be used for empty/heavy load of passenger vehicles and empty/heavy load status of trucks, manually input or obtained by sensor)
10	0: Vehicle fuel circuit normal; 1: Vehicle fuel circuit disconnected
11	0: Vehicle circuit normal; 1: Vehicle circuit disconnected
12	0: Door unlocked; 1: Door locked

13	0: Door 1 closed; 1: Door 1 open (front door)
14	0: Door 2 closed; 1: Door 2 open (middle door)
15	0: Door 3 closed; 1: Door 3 open (rear door)
16	0: Door 4 closed; 1: Door 4 open (driver door)
17	0: Door 5 closed; 1: Door 5 open (custom)
18	0: Not positioned using GPS satellite; 1: Positioned using GPS satellite
19	0: Not positioned using BeiDou satellite; 1: Positioned using BeiDou satellite
20	0: Not positioned using GLONASS satellite; 1: Positioned using GLONASS satellite
21	0: Not positioned using Galileo satellite; 1: Positioned using Galileo satellite
22-23	Reserved
24-31	X00 Invalid, not parsed; 0x10-0xFF: LockStatus (electronic lock status) (see Appendix 3 for details); Others: Reserved 0

The location additional information item format is shown in Table 26.

Table 26 Location Additional Information Item Format

Field	Data Type	Description and Requirements
Additional Information ID	BYTE	1-255
Additional Information Length	BYTE	
Additional Information		Additional information definition see Table 27

Table 27 Additional Information Definition

Additional Information ID	Additional Information Length	Description and Requirements
0x01	4	Mileage, DWORD, 1/10km, corresponding to vehicle odometer reading
0x02	2	Fuel, WORD, 1/10L, corresponding to vehicle fuel gauge reading
0x03	2	Speed recorded by driving log function, WORD, 1/10km/h
0x04	2	Event ID requiring manual confirmation of alarm, WORD, counting from 1
0x05-0x10		Reserved
0x11	1 or 5	Overspeed alarm additional information see Table 28
0x12	6	Enter/exit area/route alarm additional information see Table 29
0x13	7	Road segment driving time insufficient/too long alarm additional information see Table 30
0x14-0x24		Reserved

0x25	4	Extended vehicle signal status bit, definition see Table 31
0x2A	2	IO status bit, definition see Table 32
0x2B	4	Analog quantity, bit0-15, AD0; bit16-31, AD1
0x30	1	BYTE, wireless communication network signal strength
0x31	1	BYTE, GNSS positioning satellite count
The following are TT extensions		
0xE1	7	LBS information, see 0xE1 additional information format (used by 2G lock)
0xE2	15	IMEI code (15BYTE)
0xE3	n	Version number, ASCII string
0xE6	20	ICCID string (20BYTE)
0x5D	1+n*10	LBS information: the 1st byte is the number of base stations, followed by n base station data; base station data: see 0x5D additional information format (used by 4G lock)
0xE7	3	24-way alarm status, 3 bytes, alarm status: see 0xE7 alarm status bit definition
0xE8	3	24-way switch status; see 0xE8 switch status bit definition
0xE9	1	Device battery level, percentage
0x51	2*n	n 路温度, each 路 2 bytes, unit: 1/10 degree, 0XFFFF means temperature invalid
0x58	2*n	n 路湿度, each 路 2 bytes (表示 0.0-100.0%); e.g.: 033A means 82.6%, 0XFFF means humidity invalid
0x56	2	Device battery voltage, 2 bytes, first byte: battery percentage, 0-10, (1=>10%...10=>100%), second byte: voltage, unit 50mV, e.g. 0X50=80=>80*50mV=4000mV
0xEA	6	Total 6 bytes, G-sensor three-axis (X,Y,Z) acceleration, each axis acceleration 2 bytes, unit 1/256G, G is gravity acceleration, XYZ values use 16-bit two's complement form to distinguish positive and negative

0xE1 Package Additional Information Format (7B Base Station Information)

Starting Byte	Field	Type	Description and Requirements
0	Country Code	WORD	2-byte country code MCC, e.g. China 460=0x0460
2	Network Identifier	Byte	1-byte network identifier MNC(HEX), e.g. 12=0x0B
3	Base Station LAC	WORD	2-byte base station LAC code (HEX) e.g. 0x2842
5	Base Station	WORD	2-byte base station CELL code (HEX)

	CELL		e.g. 0x2842
--	------	--	-------------

0x5D Package Additional Information Format (10B Base Station Information)

Starting Byte	Field	Type	Description and Requirements
0	Country Code	WORD	2-byte country code MCC, e.g. China 460=0x01CC
2	Network Identifier	Byte	1-byte network identifier MNC(HEX), e.g. 12=0x0B
3	Base Station LAC	WORD	2-byte base station LAC code (HEX) e.g. 0x2842
5	Base Station CELL	DWORD	4-byte base station CELL code (HEX) 0x01501234
9	Signal Strength	Byte	1-byte signal strength

0xE7 Alarm Status Bit Definition

Bit	Status
0	1: Cable cut alarm
1	1: Emergency unlock alarm
2	1: Removal alarm
3	1: Cover opening alarm
4	1: Vibration alarm
5	1: Tilt alarm
6	1: Roll-over alarm
7	1: High temperature alarm
8	1: Low temperature alarm
9-24	Reserved, 1: parsed as 9-24 alarm

0xE8 Switch Status Bit Definition

Bit	Status
0	1: Knob opened
1	1: Motor released
2	1: Lock rod opened
3	1: Lock rod cut
4	1: Cable disconnected
5	1: Cover opened
6	1: Shell opened
7	1: Removed
8	1: Vibration, 0: Stationary
9	1: Sleep mode
10	1: Deep sleep mode
11	1: Satellite positioning off
12	1: Device power off
13	1: Firmware remote upgrading
14-19	Reserved, 1: parsed as 14-19 switch on

20-23	Network type 3210: 0:GSM; 2:UTRAN; 3:GSM W/EGPRS (above as 2G); 4:UTRAN W/HSDPA; 5:UTRAN W/HSUPA; 6:UTRAN W/HSDPA and HSUPA (above as 3G); 7:E-UTRAN; 8:UTRAN HSPA+ (above as 4G); 9:5G
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Table 28 Overspeed Alarm Additional Information Message Body Data Format

Starting Byte	Field	Data Type	Description and Requirements
0	Location Type	BYTE	0: No specific location; 1: Circular area; 2: Rectangular area; 3: Polygonal area; 4: Road segment
1	Area or Road ID	DWORD	If location type is 0, this field is absent

Table 29 Enter/Exit Area/Route Alarm Additional Information Message Body Data Format

Starting Byte	Field	Data Type	Description and Requirements
0	Location Type	BYTE	1: Circular area; 2: Rectangular area; 3: Polygonal area; 4: Route
1	Area or Route ID	DWORD	
5	Direction	BYTE	0: Enter; 1: Exit

Table 30 Road Segment Driving Time Insufficient/Too Long Alarm Additional Information Message Body Data Format

Starting Byte	Field	Data Type	Description and Requirements
0	Road Segment ID	DWORD	
4	Road Segment Driving Time	WORD	Unit: seconds (s)
6	Result	BYTE	0: Insufficient; 1: Too long

Table 32 IO Status Bit

Bit	Definition
0	1: Deep sleep state
1	1: Sleep state
2-15	Reserved

8.19 Location Information Query (Call-over)

Message ID: 0x8201.

The location information query message body is empty.

8.20 Location Information Query Response

Message ID: 0x0201.

The location information query response message body data format is shown in Table 33.

Table 33 Location Information Query Response Message Body Data Format

Starting Byte	Field	Data Type	Description and Requirements
0	Response Serial Number	WORD	Serial number of the corresponding location information query message
2	Location Information Report		Location information report see 8.12

8.23 Text Message Downlink

Message ID: 0x8300. The text message downlink message body data format is shown in Table 37.

Table 37 Text Message Downlink Message Body Data Format

Starting Byte	Field	Data Type	Description and Requirements
0	Flag	BYTE	Text message flag bit meaning see Table 38
1	Text Information	STRING	Maximum 1024 bytes, GBK encoded

Table 38 Text Message Flag Bit Meaning

Bit	Flag
0	1: Emergency
1	Reserved
2	1: Terminal display display
3	1: Terminal TTS broadcast
4	1: Advertising screen display
5	0: Center navigation information, 1: CAN fault code information
6-7	Reserved

8.61 Data Downlink Transparent Transmission

Message ID: 0x8900. The data downlink transparent transmission message body data format is shown in Table 91.

Table 91 Data Downlink Transparent Transmission Message Body Data Format

Starting Byte	Field	Data Type	Description and Requirements
0	Transparent Message Type	BYTE	0x81 Electronic lock business data
1	Transparent Message Content		See 10 TT Electronic Lock Business Protocol

8.62 Data Uplink Transparent Transmission

Message ID: 0x0900. The data uplink transparent transmission message body data format is shown in Table 92.

Table 92 Data Uplink Transparent Transmission Message Body Data Format

Starting Byte	Field	Data Type	Description and Requirements
0	Transparent Message Type	BYTE	0x81 Electronic lock business data
1	Transparent Message Content		See 10 TT Electronic Lock Business Protocol

10 TT Electronic Lock Business Protocol

10.0 Text Message Upload

Extended protocol, supplementary text message upload

Message ID = 0x0300

Data format same as 0x8300, that is: text flag + GBK text string; flag is 4

Starting Byte	Field	Data Type	Description and Requirements
0	Flag	BYTE	Flag is 4
1	Text Information	STRING	Maximum 102 bytes, GBK encoded

10.1 Electronic Lock Business Protocol Framework

(1) Communication Protocol Description:

- The formal data in this business protocol is generally hexadecimal data, unless otherwise stated
- Business data frame header "*", frame tail "#"

(2) Terminal Uplink Protocol Structure:

Data Length	Business Data	Additional Information	Business Serial Number	CRC
1B (business data to CRC length)	0x2A (business header) ... (specific business data, see below) 0x23 (business tail)	28B GPS information + 7B/10B base station information (optional, center determines whether this field exists based on data length)	2B active upload=0X0000, reply=downlink serial number	1B

(3) Center Downlink Protocol Structure:

Data Length	Business Data	Business Serial Number	CRC
1B (business data to CRC length)	0x2A (business header) (specific business data, see below) 0x23 (business tail)	2B downlink serial number, 0x0001~0XFFFF	1B

10.2 Electronic Lock Operation Command

10.2.1 Seal/Unseal/Alarm Release/Set Dynamic Password/Modify Local Password

Center Downlink:

Lock operation command (1B, 0x15) + Cmd (1B) + LockID (4B) + Gate (1B) + Bill (8B) + LineCode (2B) + Key (6B) + VaildTime (1B) + Spare (4B) + Absolute Time (6B)

252A

1532 043B1F46 02 00003039040BEB CB 04D204D20008AA52 03 011080260810164318

23 \$\$

Note: When Cmd is to set dynamic password:

Key is the latest dynamic password, VaildTime is the valid time of the dynamic password, in minutes (1~255 minutes, it is recommended not to exceed 20 minutes, if 0 it means clear dynamic password)

When Cmd is to modify local password: Key is the latest local password

Terminal Reply:

Operation reply (1B, 0x55) + Cmd (1B) + LockID (4B) + Gate (1B) + Bill (8B) + Voltage (1B) + LockStatus (1B) + MotoStatus (1B) + LineCode (2B) + Operation Identifier (8B) + cmdSRC (1B, operation source: SMS 0x00, automatic 0x01, keyboard 0x02, handheld 0x03, center 0x04, card slot 0x05, IC card 0x06, other 0x07~0x0F) + Absolute Time (6B)

Note: 1) When operating remotely, RF, keyboard:

Operation identifier (8B) = operation identifier ('R' = RF, 'C' = remote, 'K' = keyboard,) + operation command code 1B + input password 6B

2) When setting dynamic password reply:

Operation identifier (8B) = operation identifier ('1': valid dynamic password, 0: invalid dynamic password) + received dynamic password and valid time 7B

3) When operating IC card:

Operation identifier (8B) = operation identifier ('I' = IC card) + operation command code (1B, 0x60 apply card, 0x61 unseal card, 0x62 temporary unseal card) + IC card number 4B (directly HEX explicit code is enough) + remaining times 1B (applicable to temporary unseal card)

Eg: IC 2A 5597 04 3B 1F 46 02 12 34 56 78 12 34 56 78 35 20 00 05 07 15 14 48 33 23 \$\$

10.2.2 Binding Type Lock Command (7B Lock Dedicated)

10.2.2.1 Temporary Unseal Card Authorization

Downlink:

0x12 + SubCmd2 (1B, 0x02) + IC card quantity (1B) + IC card number list (4 B) + times (1B) + absolute time (6B)

Lock reply:

0x52 + SubCmd2 (1B, 0x02) + IC card quantity (1B) + IC card number list (4 B) + times (1B) + absolute time (6B)

Note: IC card quantity current fixed 0x01; times is 0 means cancel authorization, times is 255 means unlimited times, times not decremented; temporary unseal card can only unlock authorization.

10.2.2.2 Temporary Unseal Card Authorization Status Query:

Downlink:

0x12 + SubCmd2 (1B, 0x03) + absolute time (6B)

Lock reply:

0x52 + SubCmd2 (1B, 0x03) + IC card quantity (1B) + IC card number list (4 B) + remaining times (1B) + absolute time (6B)

10.2.2.3 IC Card Blacklist Operation:

Downlink:

0x12 + SubCmd2 (1B, 0x04) + subcmd3 (1B, 0x01 overwrite; 0x02 append; 0x03 delete; 0x04 clear; 0x05 query) + IC card quantity n (1B, 1~45) + IC card number list (4 B*n, when clear/query n=0, this item is absent) + absolute time (6B)

Query reply:

0x52 + SubCmd2 (1B, 0x04) + subcmd3 (1B, 0x05 query) + current blacklist total K (2B) + sequence number (1B) + IC card quantity n (1B, 1~45) + IC card number list (4 B*n) + absolute time (6B)

Note: When the IC card blacklist quantity is greater than 45, upload this reply instruction multiple times, sequence number increments.

Other replies:

0x52 + SubCmd2 (1B, 0x04) + subcmd3 (1B, 0x01 overwrite; 0x02 append; 0x03 delete; 0x04 clear) + current blacklist total K (2B) + absolute time (6B)

10.3 Electronic Lock Information Active Upload Command

Uplink:

Operation command (1B, uplink 0x01,) + SubCmd (1B) + LockID (4B) + Gate (1B) + Voltage (1B) + LockStatus (1B) + MotoStatus (1B) + Spare (4B) + Ver (1B) + absolute time (6B)

Universal reply:

Operation command (1B, reply 0x61) + SubCmd (1B) + LockID (4B) + absolute time (6B)

SubCmd:

Lock Active Upload		Center Business Reply	
Lock information upload	0x01	None	
Knob damage alarm	0x02	None	
Lock body damage alarm	0x03	None	
Lock rod cut alarm	0x04	None	
Lock rod open alarm	0x05	None	
Apply dynamic password / apply unlock	0xF2	0xF2	

Appendix 0: Protocol Related Field Description

Command Word	Description	Description	Length
Cmd	Specific operation command word	See Appendix 1: Command word list.	1B
LockID	Electronic lock number	Electronic lock number: 8 digits, respectively convert the first and last 4 digits of decimal value to 2B hexadecimal as the high and low parts, e.g. 83181001 converted to 0x207E03E9	4B (8 digits)
Gate	Channel number	Card slot channel number (e.g. channel 15 is 0x0F), linkagedownlink channel number fixed as 0x00, same below.	1B (2 digits)
Voltage	Voltage	1. Ordinary hanging lock: The corresponding voltage values of this character's hexadecimal are as follows:	1B

16 进制	Voltage Value	Description
0x30	3.3 V	Severe low voltage
0x31	3.6 V	Low voltage
0x32	3.7 V	Voltage normal
0x33	3.8V	Voltage normal
0x34	3.9V	Voltage normal
0x35	4.0V	Voltage normal
0x36	4.1V	Voltage normal
0x37	4.2 V	Full charge

2: RFID tag voltage = Voltage*3.66/256, unit V

Command Word	Description	Description	Length
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LockStatus	Lock status	Only valid for uplink information (see Appendix 3), this bit is reserved for downlink, fill 0x00	1B
Bill	Load list number	Load list number, HEX explicit code, default parsed as HEX literal value (e.g. 0x1234567890ABCDEF=1234567890ABCDEF)	8B
LineCode	Line number	Line number (convert number decimal to 2 bytes hexadecimal)	2B (4 digits)
Key	Key	As the electronic seal when sealing write into the electronic lock, when unsealing the electronic lock compare the information contained in the unsealing key with its own stored electronic seal to see if they match, then release electronic seal, otherwise fail.	6B (10 digits)
		1.RF lock: respectively convert the first 4 digits and last 6 digits of decimal value to 2B hexadecimal same below	
		2.Password lock: directly 6-bit ASCII code number	
Absolute Time	Command generation time	Hexadecimal literal meaning: e.g. 0x080402152050 is 2008-4-2 15:20:50, belongs to central time or communication local time	6B

Appendix 1: Command Word List

Center Downlink

NAME	Command Word	NAME	Command Word
Set dynamic password	0xA0	Set dynamic password success	0xA1
Modify local password	0xA2	Modify local password success	0xA3
		Seal success	0x80
		Repeat seal	0x81
		Lock not locked properly, seal failed	0x82
Seal	Remote seal: 0x32	Voltage too low, cannot seal	0x83
		Lock abnormal: illegal disassemble the housing, not sealed	0x84
		Lock abnormal: emergency unlock, not sealed	0x85
		Lock rod cut alarm, not sealed	0x86
		Lock rod open alarm, not sealed	0x87
		Seal timeout	0x89

		Unseal success	0x90
		Repeat unseal	0x91
		Lock in open state when unsealing	0x92
Unseal	Remote unseal: 0x38	Key mismatch, unseal failed	0x93
		Lock abnormal: illegal 拆壳, not unsealed	0x94
		Lock abnormal: emergency unlock, not unsealed	0x95
		Open alarm, not unsealed	0x96
		Cut alarm, not unsealed	0x98
		Unseal without sealing	0x97
		Unseal timeout	0x99
		Alarm release success	0x70
		Lock not in alarm state	0x72
Alarm release	Remote alarm release 0x42	Key mismatch, not released	0x73
		Lock abnormal: illegal open the housing, not released	0x74
		Lock abnormal: emergency unlock, not released	0x75
		Alarm release timeout	0x71

Appendix 3: Lock Status (LockStatus) Description

Comprehensive lock status, contains important alarm information

High 4 bits	Meaning (current status)	Low 4 bits
0x1x	Open state	Reserved, not parsed
0x2x	Pending state	
0x3x	Not locked properly	
0x4x	Sealed state	
0x5x	Local sealed state	
0x6x	Unsealed state	
0xBx	Local unsealed state	
0x7x	Alarm state	bit0=1: Lock rod cut alarm
0x8x	Local alarm state	bit1=1: Open alarm
0x9x	Alarm release state	bit2=1: Removal alarm

0xAx	Abnormal state	bit3=1: Knob damage alarm
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Note: When the terminal is in local sealed state, it means the terminal has been sealed. The platform needs to issue a seal command according to business needs to seal it, let the lock enter sealed state. After the platform seals, the local password becomes invalid, and the close seal password takes effect.

Appendix 5: Text Message Command

Text Command	Command Function	Remark
TT%BHVOL=DD	Low voltage protection threshold setting, DD=00~89, percentage	
TT%BHVOL=?	Query low voltage protection threshold	Text content reply
TT%OFFDTM=DDDM	delayed power-off time, DD=000~999, unit minutes	
TT%OFFDTM=?	Query Delayed Power-off Time	Text content reply
IAP1,P:112.74.131.151,19	Remote upgrade command, yellow bottom part is upgrade fileServer IP address in use, green bottom part is upgrade file name, modify as needed.	Text content reply
19,HEX,GDWgdw?104,H	(Upgrade available when battery level $\geq 70\%$, under standby / seal-released state)	Upgrade progress and result.
IAP1,P:112.74.131.151,19	Hardware unlimited upgrade command, use with caution, may cause upgrade failure leading to program loss, especially when battery is low.	Text content reply
19,HEX,GDWgdw?104,H	Whether upgrade is allowed is limited by platform.	
TT%REST=1	Static sleep function enabled	
TT%REST=0	Static sleep function disabled, factory default off	
TT%REST=?	Static sleep function switch status query	Text content reply
TT%RESTTIME=nnnnM	Static sleep start time setting, unit minutes (means continuous static nnnn minutes later enter sleep mode,nnnn>1)	7B applicable
TT%RESTTIME=?	Static sleep start time query, unit minutes	7B applicable
TT%GPS=1	Enable GPS function	7B applicable

TT%GPS=0	Disable GPS function	7B applicable
TT%VLOW=DD	Low voltage alarm threshold setting, DD=10~80, percentage	
TT%VLOW=?	Low voltage alarm threshold query	Text content reply